ECE 473 Project

October 30, 2026

Motivation

Prediction markets list contracts across a wide variety of domains, such as politics, sports, weather, crypto and macroeconomics, and each domain has its own information sources, dynamics and forecasting structure. Specialized forecasting models are well developed in some areas (sports odds, numerical weather prediction), yet prediction-market makers generally use the same market-making mechanism in every domain, despite large differences in the underlying forecasting problem. This leaves value on the table. A market maker that prices contracts from a calibrated, domain-specific forecast should quote tighter and better-calibrated prices and lose less to informed traders, improving liquidity for traders and profitability for the liquidity provider. We study this in weather markets, where contracts resolve on objective station data and a rich public forecasting ecosystem exists that general-purpose market makers ignore.

Objective

We will build a specialized market maker for sports prediction markets, where prices jump sharply on scoring plays, injuries and lineup news. Our forecasting model will combine pregame team strength ratings with a live win probability model that updates on the game state. The market maker will quote around this forecast, widening spreads during volatile moments and adjusting quotes to manage inventory. We will compare its prices and performance against a general purpose market maker quoting around the market mid, a model using only pregame odds, and vig-free odds from a sharp sportsbook.

Evaluation

We will run a historical backtest that trains on earlier seasons and tests on later ones, so no future information leaks into the model. Forecast quality will be measured with Brier score, log loss and calibration against the market price and each baseline. Market making performance will be measured by replaying historical trades to simulate fills, reporting PnL, spread earned, adverse selection and tail risk such as maximum drawdown. We will report results separately for calm periods and volatile windows around scoring plays and lineup news, since that is where a sports market maker gains or loses the most. The key result is whether our model improves out of sample calibration and PnL over the baselines. We will add theory only if our quoting rule needs a bound on worst case loss.

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Domain-Specialized Market Making for NBA Prediction Markets

ECE 473 Project One-Pager · Angel Chang Liu, Arjav Akkole, Bryan Martinez · October 7, 2026

Goal

We will build a market maker for NBA game-winner contracts on Kalshi that quotes around a calibrated live win probability, and test whether it beats a market maker quoting around the market

mid on out-of-sample calibration and PnL. Baselines: mid-quoting market maker, pregame-only model, and vig-free Pinnacle odds.

Literature Review

Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation. *R. Hanson (2007).* Hanson introduces the logarithmic market scoring rule (LMSR), the standard automated market maker for prediction markets. It prices a binary contract from the history of trades and caps the market maker's worst-case loss. It learns only from order flow; ours centers prices on an outside forecast.

High-Frequency Trading in a Limit Order Book. *M. Avellaneda and S. Stoikov (2008).* The authors derive optimal bid and ask quotes for a dealer who holds inventory. Quotes center on a reservation price that shifts away from the side the dealer is long. We adopt this structure but replace the market mid with our model's win probability.

A Brownian Motion Model for the Progress of Sports Scores. *H. S. Stern (1994).* Stern models the score difference in a game as Brownian motion with drift, using NBA data, which gives a closed-form win probability at any point in the game. This is the baseline for our live model, with the drift set by pregame team strength.

Bayesian Estimation of In-Game Home Team Win Probability for College Basketball. *J. T. Maddox, R. Sides and J. L. Harvill (2022).* The authors build Bayesian in-game win probability estimators, one of which blends in a time-weighted pregame win probability, and show they match or beat existing methods. This is the same prior-plus-live structure our forecast uses.

Makers and Takers: The Economics of the Kalshi Prediction Market. *C. Bürgi, W. Deng and K. Whelan (2025).* Studying over 300,000 Kalshi contracts, the authors find a favorite-longshot bias and that makers earn better returns than takers. This suggests a passive market maker on Kalshi can be profitable.

Evaluating Real-Time Probabilistic Forecasts with Application to National Basketball Association Outcome Prediction. *C.-K. Yeh, G. Rice and J. A. Dubin (2022).* The authors develop tools to evaluate win probability forecasts that update during a game and find ESPN's live NBA forecasts are well calibrated but no more skillful than a logistic regression on team strength and score margin. That logistic model is the bar our forecast must clear.

No prior work tests a market maker anchored to a domain forecast on NBA contracts.

Data

(1) Kalshi KXNBAGAME trades and 1-minute bid/ask candles via the free public API, 2025 playoffs through the 2026 Finals. (2) Polymarket NBA trades via its Data API, 2024–25. (3) Play-by-play with wall-clock timestamps from `nba_api`. (4) Historical Pinnacle odds from The Odds API. (5) Official NBA injury report PDFs. All data is joined on game ID and UTC timestamp.

Method

A pregame prior from Elo and the de-vigged closing line feeds a live win probability model updated each play on margin and time left. We quote YES bid and ask around this forecast, skewing for inventory and widening after scoring runs and news. The model trains on 2015–16 to 2023–24 play-by-play. We tune on the 2025 playoffs through January 2026 and test on February through

June 2026. Fills are simulated against historical trades and candles.

Schedule and Milestones

Milestone I. Scrape and match all Kalshi and Polymarket NBA game markets to game IDs (target: at least 90% matched); align play-by-play to UTC and tag scoring runs and news windows; train the prior and live model and report Brier score, log loss and calibration against the logistic benchmark and the market price; build the backtest engine and run the mid-quoting baseline.

Milestone II. Implement the forecast-centered quoting rule and add the pregame-only and Pinnacle baselines; test out of sample, reporting PnL, spread earned, adverse selection (60-second markout) and max drawdown for calm windows, volatile windows and final minutes.

Responsibilities. Angel Chang Liu: market data scrapers and matching; fill simulation and trading metrics. **Arjav Akkole:** play-by-play alignment, Elo and pregame prior; baselines. **Bryan Martinez:** live win probability model and calibration; quoting rule and tuning. **All:** backtest design, reports and presentation.